

	Success	Efficiency			Interaction quality					
	In-time	Action	Conversation	Overall	Factual grounding	Situational relevance	Actionable guidance	User intent uptake	Guidance conciseness	Overall
Assistant										
Frontier API models										
Gemini 3.8 Flash	30.6	32.7	73.7	55.1	82.7	79.6	68.1	76.2	89.0	<u>53.4</u>
GPT-6 Astra	<u>26.5</u>	33.0	<u>82.7</u>	59.9	<u>86.5</u>	89.4	78.5	97.1	<u>86.1</u>	63.5
Gemini 3.1 Pro	20.8	30.9	69.9	52.1	61.5	66.0	56.6	80.4	65.4	27.9
Real-time streaming API models										
GPT-Realtime 2.1	21.2	35.3	32.7	34.3	80.9	<u>87.7</u>	<u>72.6</u>	76.2	58.1	49.5
Gemini 3.8 Live	11.9	33.5	89.7	<u>63.2</u>	74.7	78.2	58.0	77.7	86.0	39.9
Open-weight non-streaming VLMs										
Muse Glimmer	23.8	42.5	0.8	19.6	59.2	75.3	56.8	65.0	41.3	27.3
Qwen3.8-27B	18.4	37.3	36.4	37.9	70.9	81.7	62.4	<u>85.9</u>	71.9	37.4
Kimi-VL-A3B-Thinking	15.2	30.7	24.0	27.1	34.2	39.4	29.1	39.0	26.0	5.7
InternVideo3-8B	14.5	38.0	59.7	50.2	65.7	45.3	35.0	41.7	68.9	20.9
Molmo2-8B	13.4	41.0	2.3	19.7	54.5	31.7	32.3	33.4	19.2	11.6
GLM-4.6V-Flash	12.3	38.5	82.1	63.9	90.1	58.6	54.2	62.4	82.3	47.3
Open-weight streaming VLMs										
Qwen3-Omni-30B	26.2	<u>45.0</u>	0.0	20.7	65.8	63.8	51.2	54.3	67.6	25.4
MiniCPM-o 4.5	13.7	49.5	0.0	21.3	55.4	44.7	33.5	46.5	51.8	14.6
Proact-VL	11.3	43.1	71.3	58.1	66.4	25.6	17.2	28.4	44.5	15.8
StreamingVLM	10.9	39.4	77.1	59.5	75.8	25.0	34.8	34.4	22.9	24.5

Table 1 Visual assistance in CookSim. All scores are on a 0–100 scale (↑). Success and efficiency are three-run means; efficiency is measured on in-time successes, with persona-weighted overall scores. Rubrics use run 1 and the full 16-item overall score. Bold/underline indicate the best/second-best assistant in each column; dashes denote unavailable scores.